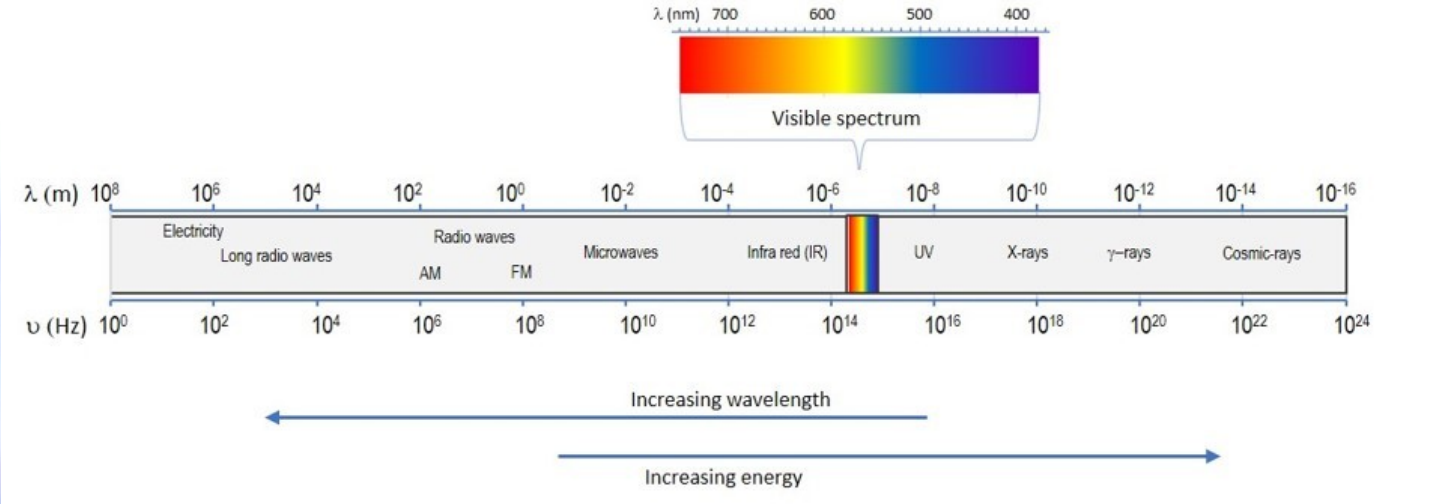
THz

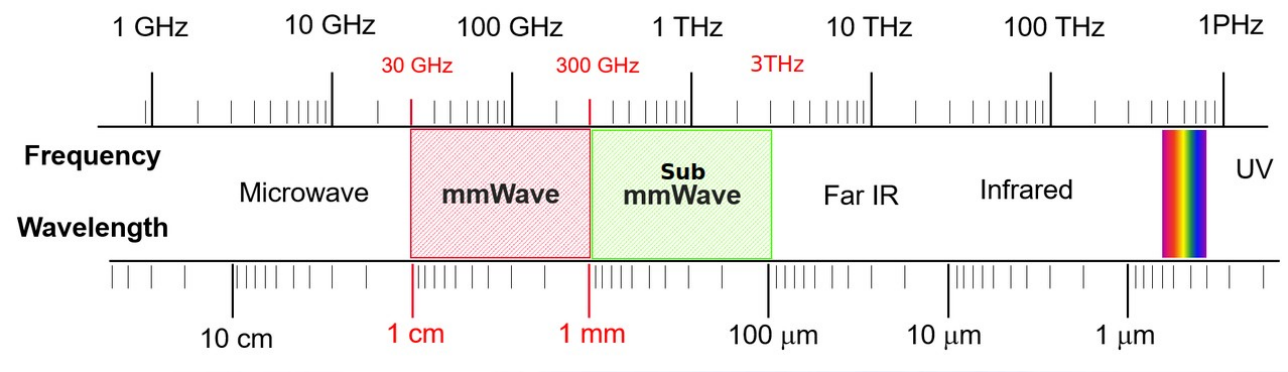
The last frontier !

MICROMEET – 2026
L. Cupido - CT1DMK

**“If you want to have good ideas,
you must have many ideas.” – Linus Pauling**

Sub-millimeter wave or THz region, 300GHz to 3THz.

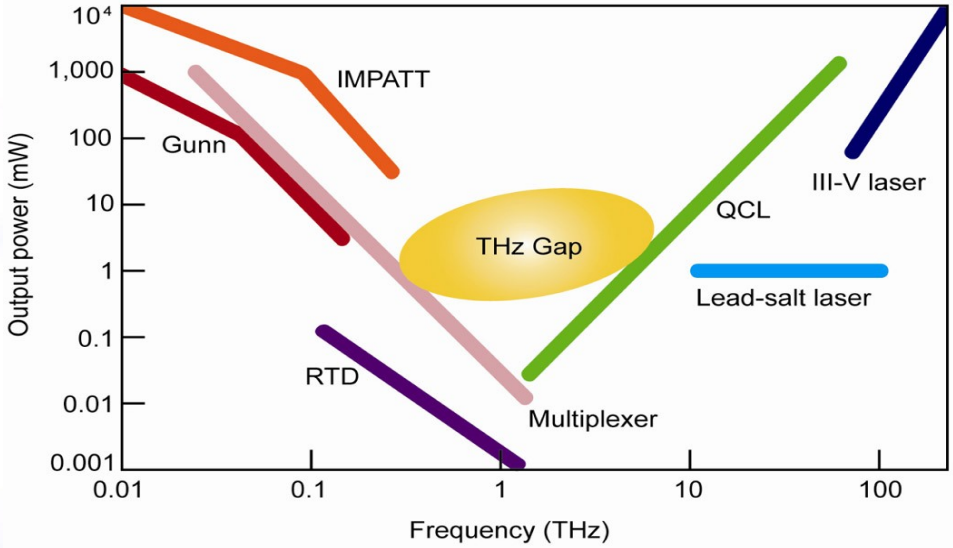




The THz Gap.

Below 300GHz generating some power is somehow easy.

-All sorts of transistors and MMIC < 50GHz...90GHz
-IMPATT diode oscillators, up to 220GHz with 10mW (10dBm)
-GUNN diode oscillators, up to 170GHz with 5mW (7dBm)
-Active Multiplier chains, up to 300GHz with 0.1mW (-10dBm)

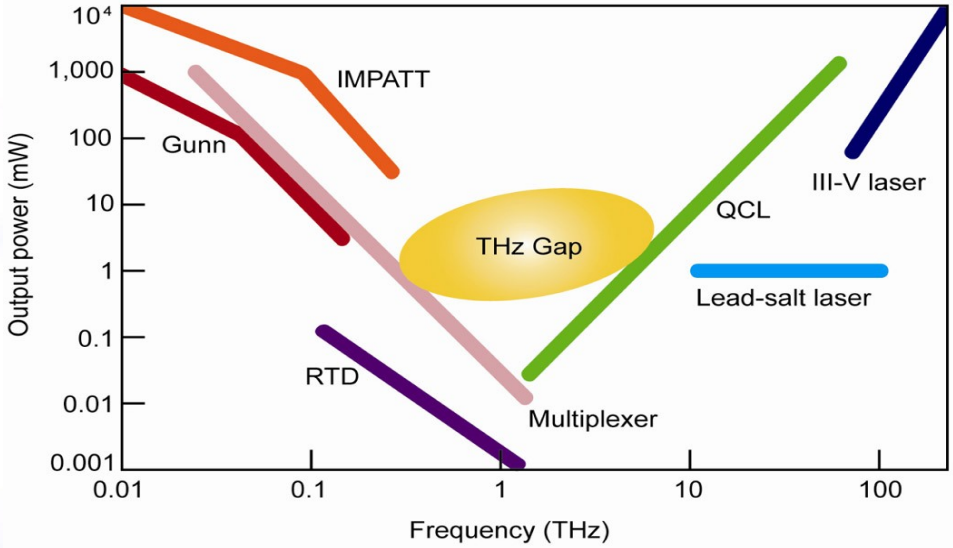


The THz Gap, the most difficult region of the EM spectrum.

Between 300GHz and 3 THz generating power is very difficult.

- Resonant tunneling diodes, 1THz with < 0.001mW (-30dBm)
- Powerful multiplier chains can reach 1.5THz with 0.01mW (-20dBm)

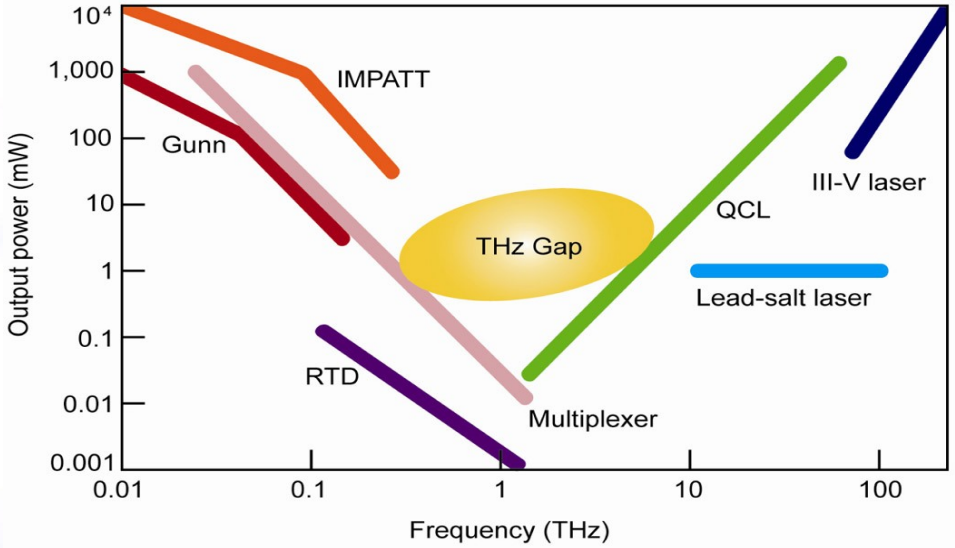
On this region we are too high for electronics and too low for photonics.



The THz Gap.

Above 3 THz generating some power becomes easy again.

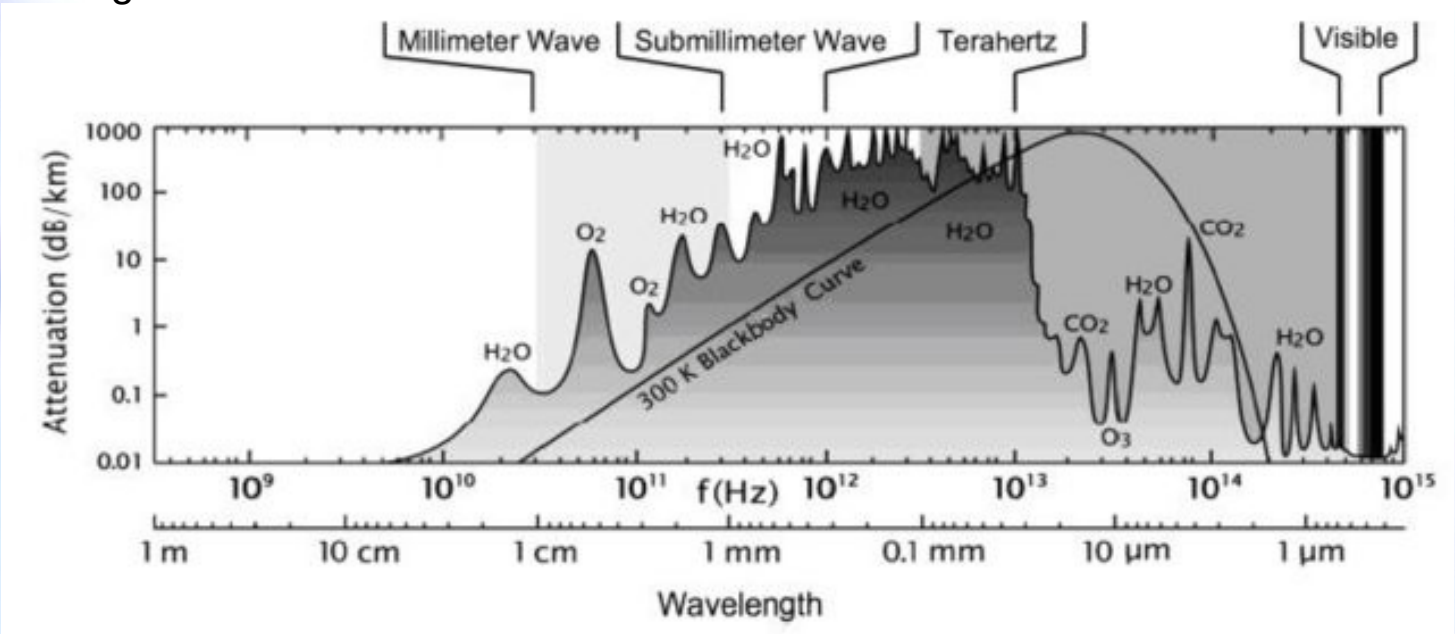
- Quantum Cascade lasers >3THz with > 1mW
- A variety of lasers above 10 THz > 10 mW
- Them above 100 THz , several Kw if necessary.



The THz Gap, the most difficult region of the EM spectrum.

For radio communications

- For short range (few meters) It has potential for tremendous bandwidth.
- For long range (> 1 Km) this is a very useless spectral region with more than 100dB/Km signal attenuation at 3THz.



- **We can use it for scientific applications** (but not for communications)
- **Or we can use it just because it is difficult** (seems a valid reason)

The THz Gap. What radio activity can we do here ?

Communications & Wireless Technology

- High-bandwidth links for next-generation wireless (beyond 5G/6G) short-range high-speed optical-like communications.

- Very very short range, Is this really useful for radio-ham communication experiments ???

Yes, in the educational perspective and subsequent bragging.

No, for real communication any 30€ Chinese made VHF hand-held radio will work 100x better.

The THz Gap. What other stuff can we do here ? (1)

- Spectroscopy & Material Characterization

Molecular fingerprinting: Solid-state characterization (Probing carrier dynamics, phonon modes, lattice vibrations, superconductors, and nanomaterials).

Biomolecules & polymers: Collective motions in proteins, hydration layers, and conformal changes.

- Ultrafast Dynamics and Time-Resolved Science

Pump-probe spectroscopy: studying carrier relaxation in materials.
Phase transitions: Tracking ultrafast changes in correlated electron systems.

- Biological, Biomedical Sensing & imaging.

Non-ionizing detection: Safe interaction with biological tissues, Water content and hydration mapping (early cancer detection)
Protein and DNA spectroscopy: Sensitive to collective low-frequency modes.

The THz Gap. What other stuff can we do here ? (2)

- Security & Non-Destructive Inspection

Concealed object detection: THz waves penetrate fabrics and plastics but are blocked by metals.

Material signatures: Identifying explosives, drugs, and hazardous chemicals.

- Plasmonics & Metamaterials

THz metamaterials: Tailored responses like negative refractive index, cloaking, filters, and tunable resonances.

Plasmonic devices: Surface plasmons at THz frequencies.

The THz Gap. What other stuff can we do here ? (3)

- Quantum and Condensed Matter Physics

Superconductors & correlated systems: THz probes energy scales relevant to superconducting gaps and collective excitations.

Topological materials: THz methods reveal dynamic responses tied to topological phases.

Quantum Hall and 2D material dynamics: Real-time probing of low-energy excitations.

- Environmental & Atmospheric Science

Remote sensing: Atmospheric gases have absorption lines in the THz region, useful for monitoring trace gases and climate-relevant species.

Astronomy and astrochemistry: THz astronomy (sub-mm observations) reveals molecular clouds, star formation regions, and interstellar medium chemistry.

The THz Gap. What other stuff can we do here ? (4)

- Fundamental Physics

Light–matter interactions: Study of nonlinear effects and transient states not accessible in other bands.

Strong-field THz physics: Exploring regimes where THz fields modify potential landscapes or electron motion.

But this is not what I want to talk about

This was just an introduction

The THz Gap. What we need to reach the THz band ?

- We need to generate THz signals

For launching a wave in to the propagation medium (RF)
For down-converting a wave into a lower frequency (LO)

All the rest is precision mechanics and “audio” frequencies.

- To generate THz signals

Gunn and **INPATT** diodes are difficult to find on the lower end of the THz range and impossible to get for the upper range.

Resonant Tunneling, not accessible at all (nor interesting)

Down **mixing** of two optical **lasers**, has too much phase noise (albeit possible if nothing else is available)

Active Multiplier Chain is the only decent option for generating a clean low phase noise signal.

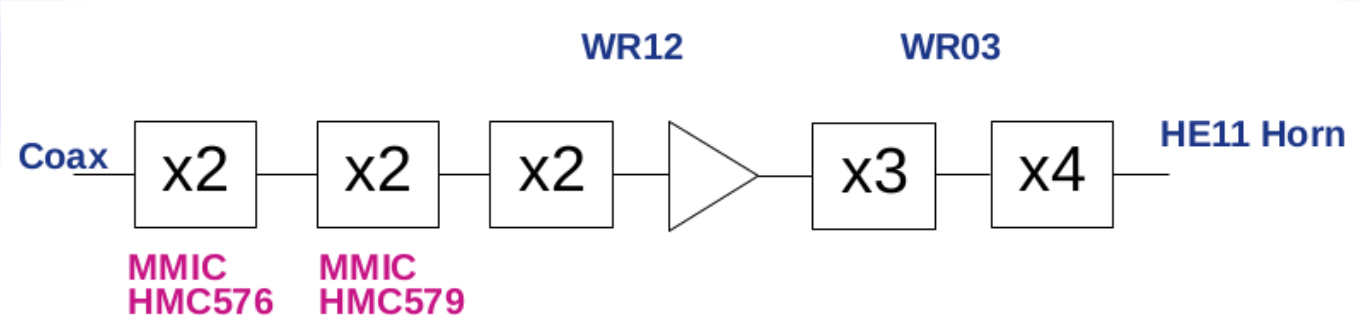
The THz Gap. Lets do a multiplier chain into the THz region.

This is just an example

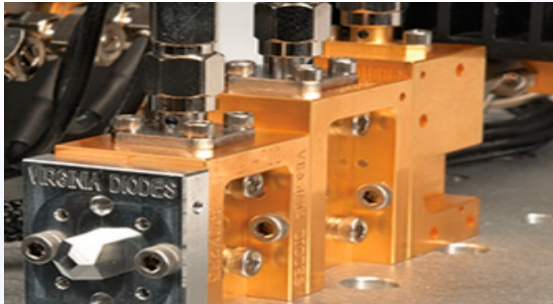
- Start in the 10-18GHz range

MMIC Active multiplier X2;	HMC576
MMIC Active multiplier x2;	HMC579
Diode doubler:	MA4E2040
Power Amplifier 80GHz:	ADPA7008
Diode trippler:	MA4E2039
Diode quadrupler:	DBES105A

for example 10.2 GHz		
>> 20.8 GHz		
>> 41.6 GHz	15 dBm	
>> 83.3 GHz	3 dBm	
>> 83.3 GHz	16 dBm	
>> 250.0 GHz	0 dBm	
>> 1.0 THz	-23 dBm	



Similar chain
By VDI.

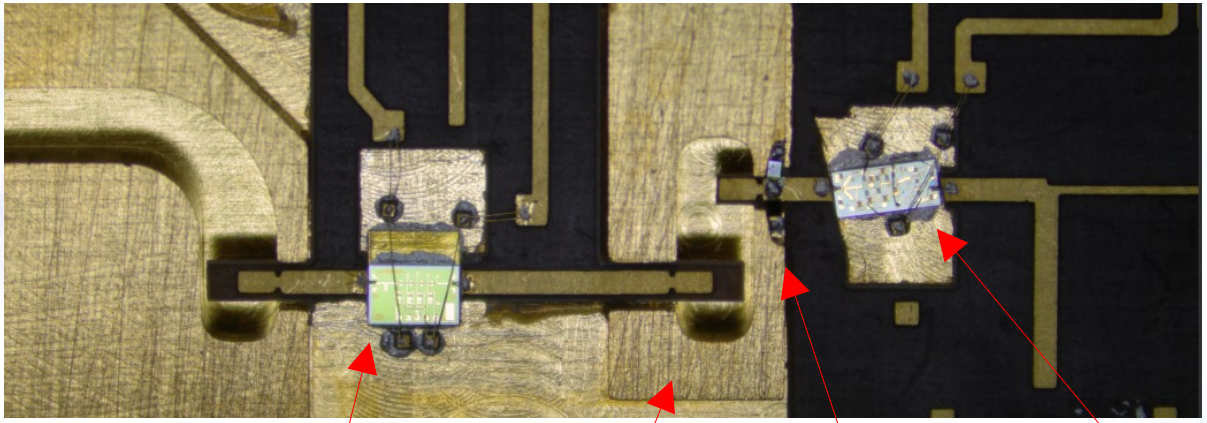


The THz Gap. Lets do a multiplier chain into the THz region.

Inside a multiplier chain to 60-90GHz *(to be followed by x2 and x4)*

Both microstrip and waveguide paths are used

Wr12 output



Amplifier

Filter/WG

X3 diodes

Active x2

By this slide I already exceeded my presentation time

The THz Gap. Circuits for wave guiding.

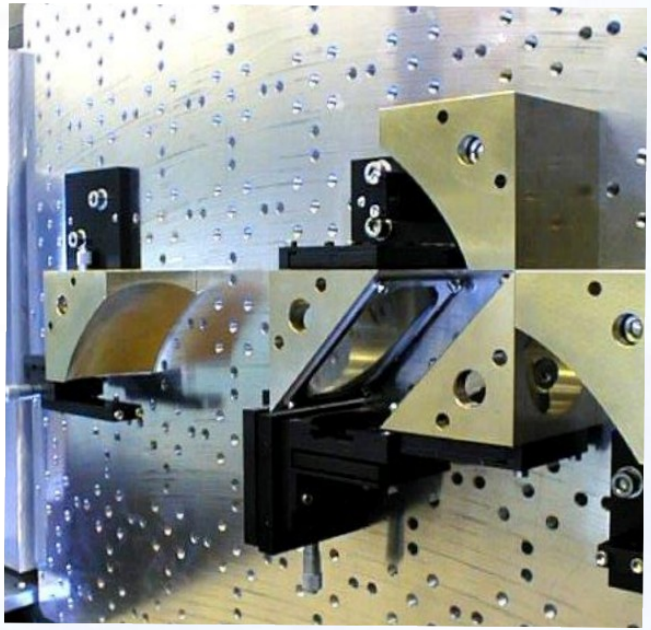
Working in fundamental wave-guide is very difficult and lossy

Open/oversized structures are quite good at THz

Guiding HE11 mode is an easy and great solution

Quasi optical setups, mirrors and lenses. (easy manuf. With CNC)

HE11 wg bends – Doty/USA



Quasi Optical combiner – TK inst./UK

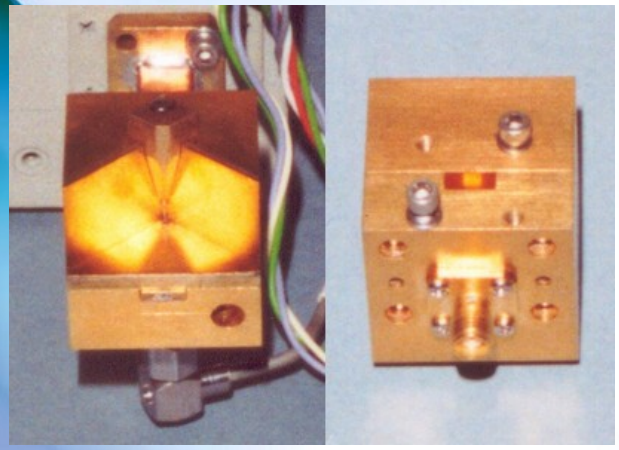
The THz Gap. Detecting and/or down-converting.

Now that we know how to make a signal we only need a mixer to downconvert it

Mixers:

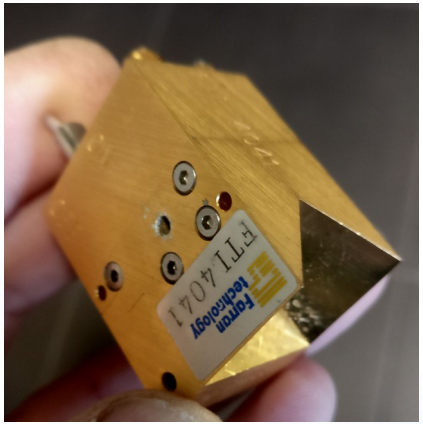
Fundamental wave-guide parts are difficult to make (too small)
Open/oversized mixers and harmonic mixers are good at THz

The infamous DMK Corner Cube

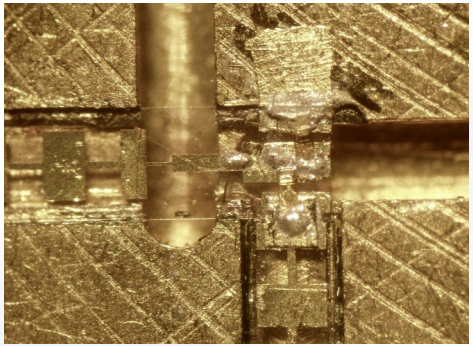


Fist QSO at 411GHz (and 725GHz tests)
By CT1DMK and CT1AMG Set. 2000

1 THz Corner cube SE mixer



Inside a fundamental
Waveguide mixer



| 1 mm |

The THz Gap.

This was just a very brief presentation about THz

There is much much more to say about THz

Many practical parts and systems can be built by hams

- requires high precision machining CNC

- modular approach is accessible to all

- fabricating modules requires wire bonding

- innovative device fabrication will require a totally different type of lab, more of a physics laboratory with sputtering machines, vacuum pumps, electron Microscope SEM, micro-lithography stuffetc.

[Be aware that there are hams out there doing really crazy stuff, a quick tour on youtube will show...]

By this slide I really exceeded by a lot so I must end the talk

THz

Is indeed the last frontier because above the THz region life becomes easy again.

Gracias

Obrigado

Thanks